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Gas Mixing Model in Complex Unconventional Tight Gas Sandstone Formations: Assessment of Long-Distance and Downward Migration

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Abstract

Accurate characterization of the fluid properties is considered one of the most important requirements in determining the potential of unconventional gas reservoirs, reducing exploration risks, predicting migration pathways, and optimizing production plans. Gas generation and migration in unconventional reservoirs are highly complex compared to conventional reservoirs due to their very low porosity and permeability. The objective of this study is to investigate the lateral and vertical fluid variation within the complex tight sandstone formations. To investigate the compositional variations of gases in unconventional reservoirs, carbon isotopic and chemical compositional analysis were carried out on gases from complex tight sandstone formations. Gas chemical composition analysis were conducted using gas-chromatography. Gases were analyzed using a gas chromatograph equipped with stable isotope ratio mass spectrometer. Carbon isotope ratios were measured for individual hydrocarbon compounds from methane to pentane, and for CO₂. The results show that the studied gases were sourced from marine source rock and are characterized by a wide range of thermal maturities. Some of the gases were sourced from a non-associated dry gas and others were sourced from oil-associated gas. The carbon isotopic compositions of carbon dioxide and the nitrogen concentration of the two endmembers are distinguishable. An isotopic fractionation model was applied to assess the possibility of mixing between the two endmembers. Data presented in this study suggest that the gases in the tight gas sandstone and shale formations represent two different source rock kitchens with different thermal maturities. Areas with distinguished carbon isotopic compositions of carbon dioxide also indicate differences within the source rock facies (i.e., the presence of calcite cement). Regions with elevated nitrogen concentrations indicate mainly a thermal decomposition of nitrogen-rich organic matter. The results also indicate a downward migration of gases from shallower source into a deeper clastic formation. The results of this study can be further integrated within the reservoir pore network and fractures/faults model to identify migration pathways and predict sweet spot location prior to drilling. These findings could contribute to reduce exploration risks and improve production plans.

Geological Background & Introduction

During the Late Ordovician, the area of interest was positioned along the margin of the Gondwana supercontinent, which experienced significant glaciation. This led to the deposition of glaciogenic formations across the plate, characterized by extensive, underfilled, valley-confined, very fine, sand-dominated deposits. In subsurface regions, however, the geological landscape is more varied, with overfilled valleys, diamictites, and unconfined, mass-emplaced, glaciomarine sandstones predominating. Following this glaciation, a significant deglaciation occurred from the latest Ordovician to early Silurian periods, transitioning to fine-grained marine shales rich in organic content (Clark-Lowes, 2005; Hayton et al., 2017; Abouelresh et al., 2020). These shales, deposited across vast areas of North Africa and the Middle East, have garnered attention as potential unconventional gas targets, particularly in light of the shale-gas boom in North America. Globally, similar organic-rich shales from the Upper Ordovician to lower Silurian are found in diverse locations, including Poland, France, the United Kingdom, and China (Hayton et al., 2017). The structural component of the studied area is very complex caused by subglacial erosion and tectonics, created accommodation space for early sedimentation and influenced depositional patterns (Clark-Lowes, 2005; Hayton et al., 2017; Meinhold et al., 2021). This relief, likely caused by subglacial erosion and tectonics, created accommodation space for early sedimentation and influenced depositional patterns, linking structural features to oceanic processes and sedimentological characteristics in the region. The base of these hot shales in Middle East reflects the Gondwanan deglaciation and subsequent transgression, with evidence of diachronous deposition noted in various regions, highlighting the complex geological history of the area (Meinhold et al., 2021). The geology of lower Paleozoic succession is a compelling subject that intertwines the region's geologic history with modern hydrocarbon exploration and extraction technologies (Clark-Lowes, 2005; Hayton et al., 2017; Meinhold et al., 2021). These sedimentary packages are recognized as challenging plays due to their generally poor reservoir qualities, which result from extensive diagenetic overprinting. Coupled with the high temperatures and pressures at these depths, the demands on drilling, completion, and stimulation activities are pushing the limits of currently available equipment and materials (Hayton et al., 2017; Abouelresh et al., 2020). Fluid type identification is a critical aspect of unconventional gas and oil exploration and development. With the rise of hydraulic fracturing and horizontal drilling, understanding the complex behavior of various fluid types in subsurface reservoir has become essential for optimizing production (Almubarak et al., 2022; Jiang et al., 2023; Alrowaie et al., 2024). To investigate the compositional variations of gases in unconventional reservoirs, carbon isotopic and chemical compositional analysis were carried out on gases from complex tight sandstone formations.

Methods

Gas Chemical Gas Isotope Analyses

In this study, gas compositional analysis and carbon isotope are used to assess the origin and the distribution of both hydrocarbons (C1-C5) and non-hydrocarbons N₂ and CO₂ in siliciclastic formations. Pressurized gas samples of siliciclastic were collected from production tests, drill stem tests or downhole samplers and analyzed for chemical composition by gas chromatography (GC) and GC isotope ratio mass spectrometer (GC IRMS). Carbon isotope analysis was measured for C1-C5 and CO₂. Two standards and multiple replicates were used and the data reproducibility is better than 0.1‰ for the standards and 0.3‰ for the samples.

Mixing Model Calculation

A linear mixing model is developed using two end-members representing two different maturities:

End Member A:

Define two or more end-members representing different hydrocarbon sources/maturities: Thermogenic dry gas (higher maturity): higher $\delta^{13}\text{C}_1$ value (-42.8‰).

End Member B: Thermogenic associated gas ** (lower maturity): Enriched in C_2+ hydrocarbons, lower $\delta^{13}\text{C}_1$ value (-50‰).

For the two end-members, the mixing ratio can be calculated using:

$$\delta^{13}\text{C}_{\text{mix}} = f \cdot \delta^{13}\text{C}_A + (1 + f) \cdot \delta^{13}\text{C}_B$$

Where:

- f = fraction of gas from end-member A,
- $\delta^{13}\text{C}_A$ and $\delta^{13}\text{C}_B$ = isotopic values of end-members.

Results and Discussion

The analysis of gas samples reveals a distinct range of chemical compositions (e.g., C_1/C_2) and carbon isotope values, which offer insights into the origin and thermal maturity of the gases (Figures). The samples exhibit wetness values ranging from 0.1 to 0.4, coupled with carbon isotope values ($\delta^{13}\text{C}_1$) between 48 and 50 ‰ for methane. This range indicates a variety of gas compositions, which can be linked to different processes of hydrocarbon formation and maturation. However, in this study, a mixing model will be used to examine if the variation of the fluid is linked to the source variation or the physical fractionation of light hydrocarbon due to fluid migration in tight sandstone formations.

Gas Origin

The wetness of the gases, defined by the ratio of heavier hydrocarbons to methane, suggests two end members (Figures 1-3). A wetness value of 0.1 indicates a gas that is predominantly dry, likely sourced from thermally mature, and high-temperature conditions. In contrast, a wetness value of 0.4 signifies a wetter gas, potentially formed from less mature organic matter (Figures 1-3). The carbon isotope values associated with methane, ranging from 48 to 50 ‰, align with typical values for thermogenic gases, which are generally produced from the thermal cracking of organic materials at elevated temperatures (Figure X). In addition, carbon isotopic values of C_{2+} show wide ranges suggesting two source rock kitchens; dry gas and associated gas (Figures 1-4).

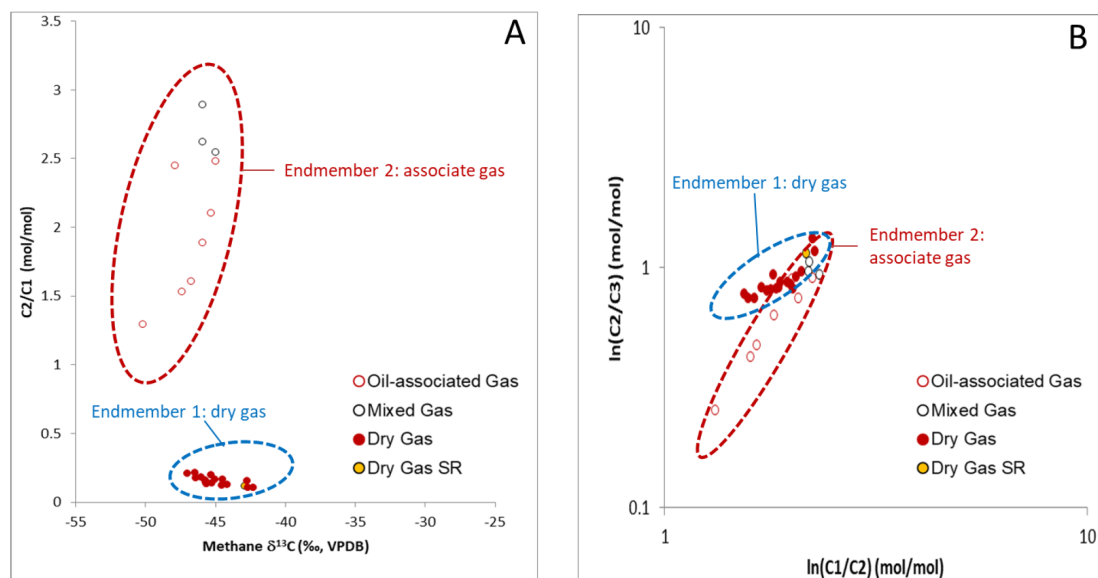


Figure 1—A) plot of carbon isotopic of methane ($\delta^{13}\text{C}$) versus the ration of ethane over methane (C_2/C_1) shows two distinctive members of source rock kitchens: dry gas (highlighted in blue) associated gas (highlighted in red). B) a plot of methane over ethane versus ethane over methane shows two distinctive clusters of two source rock kitchen with variable thermal maturity.

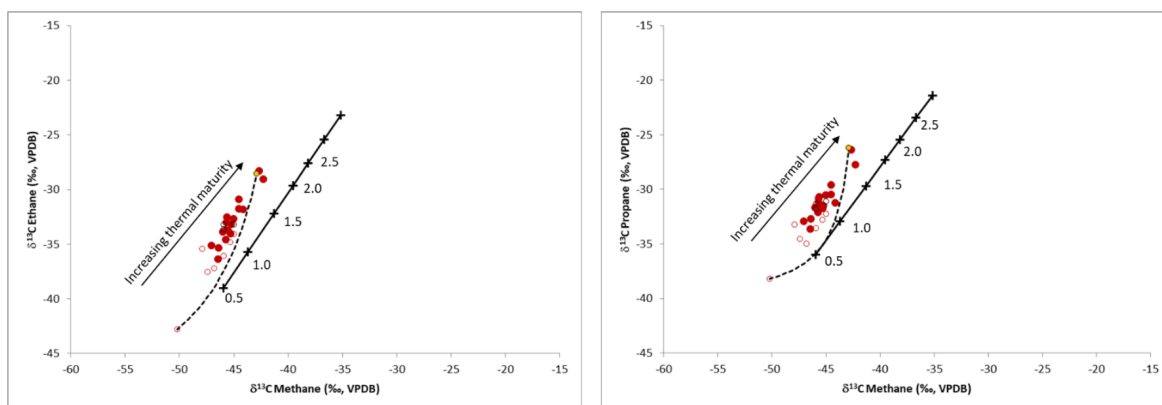


Figure 2—Plot of carbon isotopic of methane versus carbon isotopic of ethane and propane, show the thermal maturity of the studied samples. Empirical vitrinite reflectance is calculated based on [Bernard et al., 1978](#); [Chung et al., 1988](#); [Whiticar, 1999](#). The data showed that the studied gases are mainly fall within the oil-associated gas to dry gas thermal maturity level.

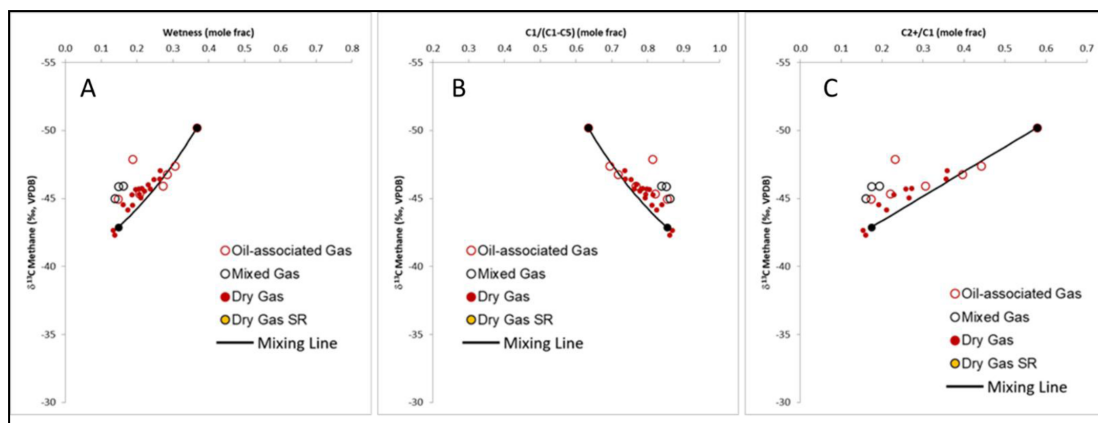


Figure 3—A) plot of wetness (C_2+ hydrocarbon/total hydrocarbons) versus carbon isotopic of methane ($\delta^{13}\text{C}$). B&C) plots of carbon isotopic of methane ($\delta^{13}\text{C}$) versus different ratios of the chemical compositions. The solid black line represents the calculated mixing line using the two endmembers.

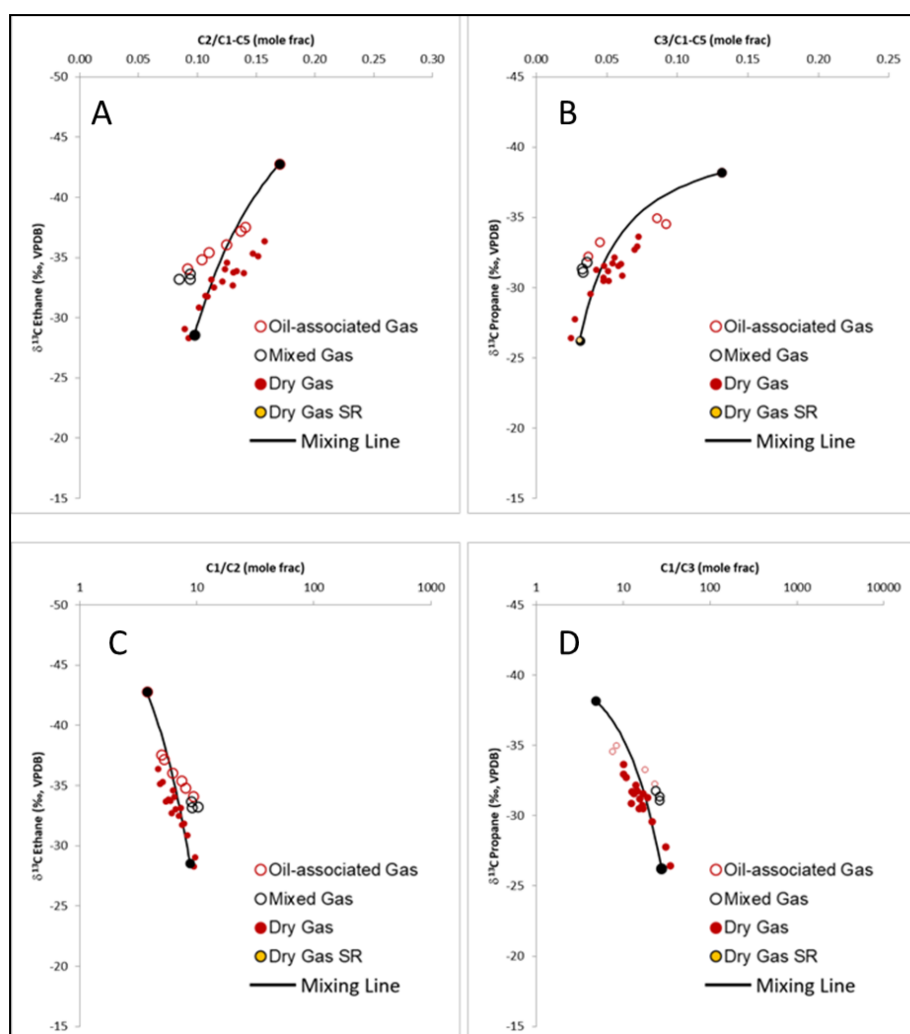


Figure 4—A-D: Plots of different gas compositions ratios and the carbon isotopic compositions of methane, ethane and propane. The plots indicate that methane over ethane and methane over propane ratios have better fit for the mixing model compared to the heavier hydrocarbons. This could suggest that the mixing was more extensive for the light hydrocarbons (i.e., methane and ethane).

Thermal Maturity

The carbon isotope composition suggests that the gases are derived from two source rock kitchens that have undergone significant thermal maturation. The observed values are consistent with gases generated in the oil window and transitioning into the gas window, indicating an associate gas to a dry gas level of thermal maturity. This is supported by the wetness values, where a dry gas composition correlates with higher thermal maturity, while a wetter gas composition may indicate varying degrees of maturity or a mixed source influence (Figures 1-2). The open black circle in the plots represent gases collected from deeper shale intervals, but with lower thermal maturities compared to the shallower gases. This could suggest that these gases represent a mixed maturities between the source rock kitchens.

The presence of carbon isotopes in the specified range further suggests that the gases are likely sourced from organic-rich source rocks that have been subjected to extensive heating over geological timescales. The values also imply that the gases may have migrated from a marine source rock, where the thermal conditions were more conducive to gas generation (Figures 1-3). However, the variation in the thermal maturities, indicate two groups of thermal maturities levels and some of the studied gases are falling in between these two groups (Figures 1-4).

Gas Mixing Model

A linear mixing model that employs gas chemical and isotopic compositions indicates the presence of mixing between two hydrocarbon members, as illustrated in the figures. The dashed line in the diagrams represents the mixing model between dry gas and associated (wet gas). Notably, data points for C2+ hydrocarbons align more closely with the mixing line, suggesting that mixing is particularly pronounced within these heavier hydrocarbons. In contrast, methane data exhibit better fit on the mixing model and tend to plot close from the calculated mixing line. This observation may imply that methane has been generated from multiple kitchens with differing thermal maturities and mixed vertically and laterally via migration indicating a complex geological history.

One possible explanation for the scattering of C3+ hydrocarbons is that methane has migrated and mixed more rapidly through the small pores present in tight sandstone formations, leading to more physical fractionation in the heavier hydrocarbons (Figures 3 and 4). Additionally, the mixing line indicates that some gases may have traveled through different intervals, highlighting the potential for both upward and downward migration pathways within the geological framework.

Global examples of mixing between dry gas and associated gas are evident in several prominent unconventional gas formations. In the Marcellus Shale, studies have documented the mixing of dry and wet gas constituents, with analyses of gas samples from various depths revealing that deeper samples yield drier gas while shallower samples contain higher concentrations of heavier hydrocarbons (Boyer et al., 2012). This variability is attributed to multiple source kitchens and migration pathways, reflecting complex geological processes (He et al., 2015). Similarly, the Barnett Shale exhibits evidence of mixing, as research indicates that gases from different intervals show variations in isotopic signatures, suggesting contributions from multiple source rocks (Sattler et al., 2015). The mixed gas composition in this formation is influenced by migration processes, where lighter hydrocarbons migrate more quickly, leading to fractionation effects (Mack et al., 2016). The Vaca Muerta Formation in Argentina also showcases this mixing, with geochemical analyses revealing varying C2+ hydrocarbon content produced from different sections, indicating contributions from both thermogenic and biogenic sources (González et al., 2018). This mixing has significant implications for understanding reservoir behavior and optimizing production strategies. Lastly, in the Flekkefjord Formation, studies have highlighted the complexity of hydrocarbon compositions, with isotopic analyses indicating migration from different source intervals and varying thermal maturities influencing gas composition (Dahl et al., 2019). The spatial variability in gas properties across these formations underscores the importance of migration pathways in shaping overall gas profiles.

The open black circles in the diagram represent gases originating from deeper formations, which are plotted within the middle maturity level. This positioning indicates that these gases exhibit characteristics of mixed origins, sourced from both dry gas thermogenic and associated gas endmembers. Such a mixture is often observed in unconventional reservoirs, where various hydrocarbon generation processes can contribute to the overall gas composition. $\delta^{13}\text{C}_1$ (methane) values shows systematic trends (i.e., increase in isotopic ratios) for both end members, when looking at special distribution. Similar trends are observed for the ethane and the propane isotopic ratios. The systematic increase in the isotopic values is related to maturity trends. The carbon isotopic values increase with increasing the thermal maturity. The systematic increase in the thermal maturity of gases can be used to indicate the migration pathways; assuming that the more mature gases are closer to the source. Both end members show systematic special increase in the thermal maturity.

Downward migration of hydrocarbons, particularly through micro-fractures, plays a crucial role in the dynamic behavior of unconventional resources. Micro-fractures, which are small-scale fractures within rock formations, provide pathways for gas and other fluids to migrate from deeper, thermally mature formations upward or laterally into shallower reservoirs. This process can be particularly significant in regions where there are overlapping geological formations with varying thermal histories. For example, in the Barnett

Shale of Texas, studies have shown that gases can migrate downward from deeper formations, such as the Ellenburger Group, into the Barnett Shale (Sharma et al., 2014; Sattler et al., 2015). The complex fracture networks within these unconventional reservoirs facilitate the movement of gases, allowing for the mixing of deeper thermogenic gases with shallower biogenic sources. This mixed gas composition can enhance the overall productivity of the reservoir, as it combines the high-quality thermogenic gas with the lighter components from shallower sources (Mack et al., 2016). Another example can be found in the Marcellus Shale, where downward migration has been observed from underlying formations like the Utica Shale (Boyer et al., 2012). The micro-fractures within the Marcellus create pathways that allow for the influx of gases from the Utica, contributing to the mixed gas characteristics noted in production wells (He et al., 2015). Additionally, in the Haynesville Shale, research indicates that downward migration pathways can lead to the accumulation of gas within the shale, as thermogenic gases from deeper formations travel through micro-fractures (Ruppel, 2011). This migration not only influences the gas composition but also impacts the pressure conditions within the reservoir, which can be critical for successful extraction.

Origin of Non-Hydrocarbons

The results of non-hydrocarbon concentration also indicate the presence of two endmembers of gas sources (two different kitchens). The results of the non-hydrocarbons show two distinguishable clusters between dry gas (low N_2 , high $\delta^{13}C_{CO_2}$ values) and associated gas (high N_2 and low $\delta^{13}C_{CO_2}$ values) (Figure).

Organically derived CO_2 generates more negative $\delta^{13}C_{CO_2}$ values (less than -10‰) in comparison to $\delta^{13}C_{CO_2}$ values from inorganic sources, such as calcium carbonate ($CaCO_3$) (Andresen et al., 1995; Dai et al., 1996; Golding et al., 2013). In this study, the observed $\delta^{13}C_{CO_2}$ values indicate that the CO_2 originates from both organic and inorganic sources, as illustrated in Figure X. The dry gas samples are characterized by enriched ^{13}C (higher $\delta^{13}C_{CO_2}$ values), ranging from -3‰ to $+1\text{‰}$, suggesting that these gases primarily derive from inorganic sources, particularly through the decomposition of calcium carbonate within the rock formation. Conversely, the associated gas samples exhibit lower $\delta^{13}C_{CO_2}$ values, indicative of an organic source originating mainly from the thermal decomposition of organic matter, such as hydrocarbons. Additionally, the black open circles representing deeper gas samples show characteristics of a mixed origin, combining both inorganic and organic sources (Figure X). This duality in origin reflects the complex interactions and processes occurring within the geological formations, providing insights into the sources of CO_2 in this context. Global examples of unconventional resources containing both organically derived and inorganic CO_2 include the Barnett Shale in Texas, where CO_2 from organic thermal decomposition mixes with inorganic sources from carbonate dissolution (Sattler et al., 2015; Mack et al., 2016). The Vaca Muerta Formation in Argentina also showcases this mixing, with CO_2 from kerogen breakdown alongside carbonate mineral CO_2 (González et al., 2018). Similarly, in the Marcellus Shale, varying $\delta^{13}C_{CO_2}$ values indicate both organic and inorganic contributions (He et al., 2015; Boyer et al., 2012). The Flekkefjord Formation in Norway presents a similar duality in CO_2 sources (Dahl et al., 2019).

The high nitrogen concentrations, exceeding 3 mol. %, observed in the studied gas samples are notably found in the associated gas samples, as illustrated in Figure 5. This elevated nitrogen content is likely linked to the decomposition of organic matter, a process that releases various gases, including nitrogen, during the thermal maturation of organic-rich sediments. The correlation between high nitrogen levels and low carbon isotopic values of CO_2 further supports this hypothesis, indicating that the nitrogen is derived from biogenic sources related to organic material breakdown (Figure 5).

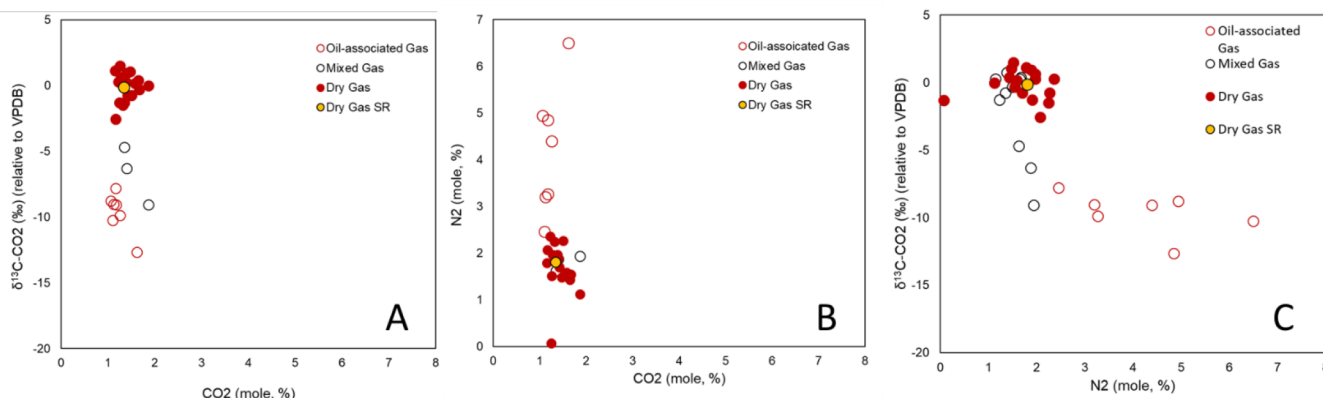


Figure 5—A) plot of carbon dioxide concentration versus the carbon isotopic composition of carbon dioxide ($\delta^{13}\text{C}\text{CO}_2$) shows two distinctive origins of CO_2 . The high carbon isotopic values, more than -3‰ , for the dry gas samples indicate that CO_2 was mainly sourced from inorganic, mainly from the decomposition of the carbonate minerals. **B)** Plot of carbon dioxide concentration versus nitrogen concentration. The carbon dioxide concentration has small variation compared to the nitrogen. **C)** Plot of nitrogen concentration versus the carbon isotopic composition of carbon dioxide. The high nitrogen concentration of nitrogen correlated with the low values of $\delta^{13}\text{C}\text{CO}_2$, indicating that the high nitrogen was mainly sourced from the decomposition of organic matter.

Additionally, the data demonstrates a lack of correlation between nitrogen concentrations and the levels of helium and CO_2 , suggesting that the high nitrogen content cannot be attributed to interactions with minerals or metasedimentary rocks in the surrounding geological formations (Figure 6). This finding is significant as it implies that the nitrogen is not a result of the typical processes associated with gas generation in sedimentary environments, such as the dissolution of nitrogen-bearing minerals or the release of gases during mineral weathering. Instead, the data strongly points toward the conclusion that the high nitrogen concentrations in the associated gas samples are predominantly a byproduct of organic matter decomposition, reinforcing the understanding of the gas composition and its origins within the studied geological context. The implications of these findings are crucial for interpreting the reservoir characteristics and optimizing extraction strategies, particularly in unconventional gas plays where understanding gas composition is vital for enhancing production efficiency.

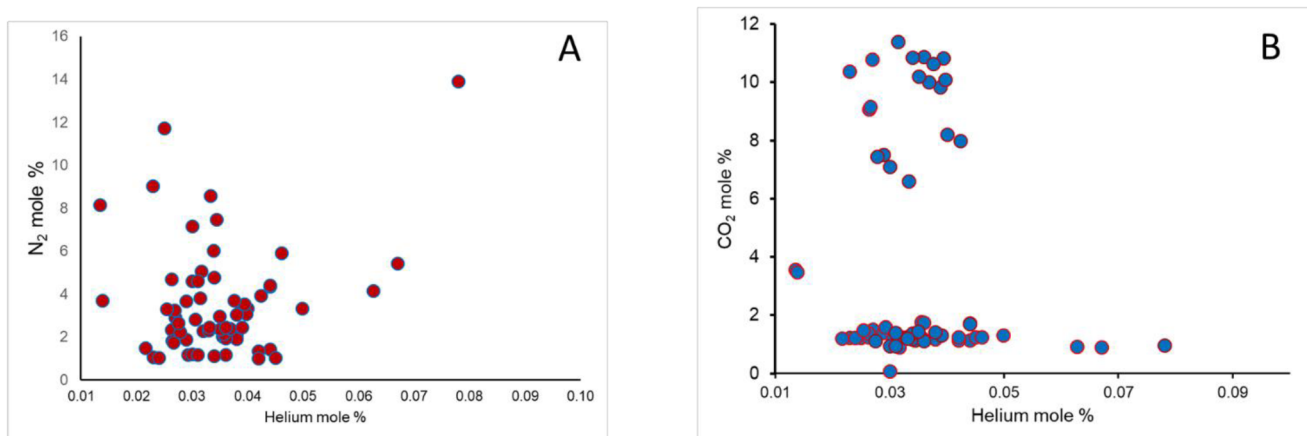


Figure 6—A) plot of helium concentration versus nitrogen concentration shows that the high nitrogen is mainly associated with low helium concentration, indicating an organic nitrogen source. **B)** Plot of helium concentration versus carbon dioxide concentration shows the lack of positive correlation between the high CO_2 with helium, eliminating the deep source of CO_2 .

High nitrogen concentrations in unconventional gas resources can indicate specific geological processes and conditions. In the Barnett Shale, elevated nitrogen levels often stem from the thermal degradation of organic matter, where nitrogen-rich compounds are released during kerogen maturation, as well as from interactions with nitrogen-rich minerals (Sattler et al., 2015). Similarly, in the Haynesville Shale,

nitrogen may originate from migration through formations containing nitrogen-bearing minerals, as well as from mixing of gases from multiple source kitchens (Mack et al., 2016). The Vaca Muerta Formation in Argentina also exhibits high nitrogen levels, likely resulting from organic matter breakdown and interactions with surrounding sedimentary layers (González et al., 2018). Additionally, in Brazil's Acreage Shale, elevated nitrogen concentrations may arise from organic-rich source rocks releasing nitrogen during thermal maturation, along with accumulation from surrounding strata (Lima et al., 2020). These examples illustrate that high nitrogen levels can result from a combination of organic matter degradation, mineral interactions, and geological complexities, impacting gas composition and production strategies.

Conclusion

In conclusion, this study provides a comprehensive analysis of fluid properties within complex tight sandstone formations, highlighting the critical importance of accurate characterization for effective unconventional gas reservoir development. The results reveal significant lateral and vertical variations in gas composition, linked to multiple source rock kitchens with differing thermal maturities, which are crucial for understanding gas generation and migration dynamics. The application of carbon isotopic and chemical compositional analyses has demonstrated distinct clusters of gas types, indicating interactions between thermogenic dry gas and oil-associated gas. Furthermore, the notable presence of high nitrogen concentrations, attributed primarily to organic matter decomposition, underscores the complex geological processes at play. These findings not only enhance the understanding of gas composition and its origins but also provide valuable insights for optimizing production strategies and reducing exploration risks. By integrating these results within reservoir models, the study contributes to identifying migration pathways and potential sweet spots, ultimately facilitating more efficient extraction techniques in unconventional gas plays. The outcomes of this research pave the way for future investigations into the intricate relationships between geological history, fluid properties, and hydrocarbon potential, further advancing the field of unconventional resource exploration.

References

- Abouelresh, M., Babalola, L., Bokhari, A., Omer, M., Koithan, T., & Boyde, D. (2020). Sedimentology, geochemistry and reservoir potential of the organic-rich Qusaiba Shale, Tabuk Basin, NW Saudi Arabia. *Marine and Petroleum Geology*, **111**, 240–260.
- Almubarak, Z., Alrowaie, M., Lu, F., & Algeer, R. (2022, September). Gas Chemical and Carbon Isotope Composition as a Diagnostic Tool for Energy. In SPE Annual Technical Conference and Exhibition? (p. D011S011R003). SPE.
- Alrowaie, M. A., Lu, F. H., Mahmoud, D. I., & Sequeira, D. S. (2024, February). A Novel Isotopic Technique to Distinguish Nitrogen Source in Downhole Sampling of Natural Gas. In International Petroleum Technology Conference (pp. IPTC-23484). IPTC.
- Bernard, B., et al. (1978) – Light hydrocarbons in recent Texas continental shelf and slope sediments. *JGR*, **83**(C8), 4053–4061.
- Boyer, C. M., Hennings, P. H., & Brantley, S. L. (2012). "Hydrocarbon migration pathways in the Marcellus Shale." *AAPG Bulletin*.
- Chung, H.M., et al. (1988) – Origin of gaseous hydrocarbons in subsurface environments: Theoretical considerations of carbon isotope distribution. *Chemical Geology*, **71**(1-3), 977–103.
- Clark-Lowes, D. D. (2005). Arabian glacial deposits: recognition of palaeovalleys within the upper Ordovician Sarah formation, Al Qasim district, Saudi Arabia. *Proceedings of the Geologists' Association*, **116**(3-4), 331–347.
- Dahl, M., H. M., A. M. S., & H. T. (2019). "Geochemical characterization of the Flekkefjord Formation." *Journal of Petroleum Geology*.
- González, A., & L. M. (2018). "Geochemical analysis of hydrocarbons in the Vaca Muerta Formation." *Energy & Fuels*.
- González, A., & L. M. (2018). "Geochemical analysis of hydrocarbons in the Vaca Muerta Formation." *Energy & Fuels*.
- Hayton, S., Rees, A. J., & Vecoli, M. (2017). A punctuated Late Ordovician and early Silurian deglaciation and transgression: Evidence from the subsurface of northern Saudi Arabia. *AAPG Bulletin*, **101**(6), 863–886.
- He, J., Heller, P. L., & Dusseault, M. B. (2015). "Characterization of gas migration in the Marcellus Shale." *Journal of Petroleum Science and Engineering*.

- Jiang, C., Wang, G., Song, L., Huang, L., Wang, S., Zhang, Y., Huang, Y., Dai, Q., & Fan, X. (2023). Identification of fluid types and their implications for petroleum exploration in the shale oil reservoir: A case study of the Fengcheng Formation in the Mahu Sag, Junggar Basin, Northwest China. *Marine and Petroleum Geology*, **147**, 105996.
- Lima, F. C., & A. R. (2020). "Geochemical characteristics of gas in the Acreage Shale." *Brazilian Journal of Petroleum and Gas*.
- Mack, L. E., & Dutton, A. (2016). "Gas composition and source rock correlations in the Barnett Shale." *Energy & Fuels*.
- Mack, L. E., & Dutton, A. (2016). "Gas composition and source rock correlations in the Barnett Shale." *Energy & Fuels*.
- Meinhold, G., Bassis, A., Hinderer, M., Lewin, A., & Berndt, J. (2021). Detrital zircon provenance of north Gondwana Paleozoic sandstones from Saudi Arabia. *Geological Magazine*, **158**(3), 442–458.
- Ruppel, S. C. (2011). "Gas shale resources in the Haynesville Shale." *U.S. Geological Survey*.
- Sattler, K., O'Brien, M., & Sweeney, M. (2015). "The role of micro-fractures in hydrocarbon migration in unconventional reservoirs." *Journal of Natural Gas Science and Engineering*.
- Sattler, K., O'Brien, M., & Sweeney, M. (2015). "The role of micro-fractures in hydrocarbon migration in unconventional reservoirs." *Journal of Natural Gas Science and Engineering*.
- Sharma, R., Kumar, S., & Tiwari, A. (2014). "Downward migration in the Barnett Shale." *International Journal of Petroleum Science and Technology*.
- Whiticar, M.J. (1999) – Carbon and hydrogen isotope systematics of bacterial formation and oxidation of methane. *Chemical Geology*, **161**(1-3), 291–314.